THREE UPPER BOUNDS FOR THE COHOMOLOGICAL DIMENSION OF A FINITELY GENERATED IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The explicit description of the derived sections with support in the closed subset of a finitely generated ideal by the Cech type complex on the generators. Invoked at the start of the counted proof (source lines 705-706) as a cross-chapter pointer. Shows directly that the number of generators bounds the cohomological dimension.
- The independence of the derived sections with support of the chosen finitely generated ideal defining the closed subset, together with the vanishing of the cohomology of a quasi-coherent sheaf on a space covered by a bounded number of affines. Both are invoked inside the counted proof (source lines 707-715 and 716-718) as cross-chapter pointers. The first transfers the bound from one set of generators to any other, the second converts an affine covering bound into a vanishing bound.

Source reference. Original reference dualizing-lemma-local-cohomology-adjoint: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 335–335. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-local-cohomology-adjoint, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference dualizing-section-local-cohomology: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 241–241. Navigation pointer only: the counted text refers to the Stacks item dualizing-section-local-cohomology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged

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Source reference. Original reference more-algebra-lemma-local-cohomology-closed: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 714–714. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-local-cohomology-closed, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-local-cohomology-is-local-cohomology: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 405–405. Navigation pointer only: the counted text refers to the Stacks item lemma-local-cohomology-is-local-cohomology, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-cd: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 679–679. Navigation pointer only: the counted text refers to the Stacks item lemma-cd, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-vanishing-nr-affines: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 718–718. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-vanishing-nr-affines, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $I \subset A$ be a finitely generated ideal of a ring A. Then*

(1) *$cd(A, I)$ is at most equal to the number of generators of I,*
(2) *$cd(A, I) \leq r$ if there exist $f_1, \ldots, f_r \in A$ such that $V(f_1, \ldots, f_r) = V(I)$,*
(3) *$cd(A, I) \leq c$ if $\mathrm{Spec}(A) \setminus V(I)$ can be covered by c affine opens.*

Proof. The explicit description for $R\Gamma_Y(-)$ given in Dualizing Complexes, Lemma dualizing-lemma-local-cohomology-adjoint shows that (1) is true. We can deduce (2) from (1) using the fact that $R\Gamma_Z$ depends only on the closed subset Z and not on the choice of the finitely generated ideal $I \subset A$ with $V(I) = Z$. This follows either from the construction of local cohomology in Dualizing Complexes, Section dualizing-section-local-cohomology combined with More on Algebra, Lemma more-algebra-lemma-local-cohomology-closed or it follows from Lemma lemma-local-cohomology-is-local-cohomology. To see (3) we use Lemma lemma-cd and the vanishing result of Cohomology of Schemes, Lemma coherent-lemma-vanishing-nr-affines. □